EXERCISE SET 12.3 — SOLUTIONS

Class 9 Mathematics | Chapter: Quadrilaterals

Question 1

(i) If P, Q, R are the midpoints of sides AB, AC, BC respectively of ΔABC, show that ΔPQR is congruent to ΔQPA and to two other triangles which you should identify.
(ii) Suppose someone erases ΔABC, leaving only ΔPQR on the paper. Can you reconstruct ΔABC from ΔPQR?

> ***Midpoint Theorem:*** *The line segment joining the midpoints of two sides of a triangle is parallel to the third side and is half as long.*

SOLUTION:

(i) Proof of Congruence:

1. Since P and Q are midpoints of AB and AC, by Midpoint Theorem:
   $PQ \parallel BC$ and $PQ = \frac{1}{2} BC = BR = RC$.
2. Since P and R are midpoints of AB and BC:
   $PR \parallel AC$ and $PR = \frac{1}{2} AC = AQ = QC$.
3. Since Q and R are midpoints of AC and BC:
   $QR \parallel AB$ and $QR = \frac{1}{2} AB = AP = PB$.

Now, comparing ΔPQR and ΔQPA:

- $PQ = QP$ (Common side)
- $PR = AQ$ (From step 2)
- $QR = AP$ (From step 3)

By **SSS Congruence Criteria**: $\Delta PQR \cong \Delta QPA$.

The other two triangles are ΔPBR and ΔRQC.

- Similarly, $\Delta PQR \cong \Delta PBR$ (by SSS criteria).
- And $\Delta PQR \cong \Delta RQC$ (by SSS criteria).

(ii) Reconstruction of ΔABC:

Yes, we can reconstruct ΔABC from ΔPQR using the following steps:

1. Draw a line passing through vertex P parallel to side QR.
2. Draw a line passing through vertex Q parallel to side PR.
3. Draw a line passing through vertex R parallel to side PQ.
4. The three points of intersection of these lines give the original vertices A, B, and C of ΔABC.

Question 2

In ΔABC, let M and N be midpoints of AB and AC respectively. Let D be any point on BC. Show that MN bisects AD.

> ***Converse of Midpoint Theorem:*** *A line drawn through the midpoint of one side of a triangle parallel to another side bisects the third side.*

SOLUTION:

1. Let AD intersect segment MN at point O.
2. In ΔABC, since M and N are midpoints of AB and AC, $MN \parallel BC$ (by Midpoint Theorem).
3. Now consider ΔABD:
   - M is the midpoint of AB.
   - MO is parallel to BD (since MO is a part of MN and $MN \parallel BC$).
4. By the **Converse of Midpoint Theorem** in ΔABD, line MO bisects AD.
5. Therefore, point O is the midpoint of AD, proving that MN bisects AD.

Question 3

In a quadrilateral $ABCD$, suppose $AB \parallel DC$ and $AB \neq CD$. Suppose G and H are the midpoints of AC and BD respectively. Prove that $GH \parallel AB$. (Why did we assume $AB \neq CD$?)

SOLUTION:

1. Let M be the midpoint of side AD.
2. In ΔADC: M is the midpoint of AD and G is the midpoint of AC.
   By Midpoint Theorem: $MG \parallel CD$ and $MG = \frac{1}{2} CD$.
3. In ΔDAB: M is the midpoint of AD and H is the midpoint of BD.
   By Midpoint Theorem: $MH \parallel AB$ and $MH = \frac{1}{2} AB$.
4. Since given $AB \parallel CD$, it follows that $MG \parallel AB$ and $MH \parallel AB$.
5. By Euclid's Parallel Postulate, points M, H, and G must be collinear (lie on the exact same straight line).
6. Since line MHG is parallel to AB, it directly follows that $GH \parallel AB$.

Reason for assumption $AB \neq CD$:

- If $AB = CD$ (given $AB \parallel CD$), $ABCD$ would be a parallelogram.
- In a parallelogram, diagonals bisect each other, which means the midpoints G and H would coincide at the same exact point ($G = H$).
- If G and H are the same point, segment GH does not exist as a line segment. Hence, $AB \neq CD$ ensures G and H are distinct points.

Question 4

Suppose the midpoints of sides AB, BC, CD and DA of a quadrilateral $ABCD$ are P, Q, R and S respectively.
(i) Show that PR and QS bisect each other.
(ii) Show that if $AC = BD$, then PR and QS are perpendicular. Is the converse true?

SOLUTION:

(i) Proof that PR and QS bisect each other:

1. Join diagonal AC.
2. In ΔABC, P and Q are midpoints of AB and $BC \Rightarrow PQ \parallel AC$ and $PQ = \frac{1}{2} AC$.
3. In ΔADC, S and R are midpoints of AD and $CD \Rightarrow SR \parallel AC$ and $SR = \frac{1}{2} AC$.
4. Since $PQ \parallel SR$ and $PQ = SR$, quadrilateral $PQRS$ **is a parallelogram** (Varignon Parallelogram).
5. PR and QS are the diagonals of parallelogram $PQRS$.
6. Since diagonals of a parallelogram bisect each other, PR and QS bisect each other.

(ii) Proof when $AC = BD$:

1. In ΔABD, P and S are midpoints $\Rightarrow PS = \frac{1}{2} BD$.
2. From part (i), $PQ = \frac{1}{2} AC$.
3. Since given $AC = BD$, we get $PQ = PS$.
4. Parallelogram $PQRS$ with equal adjacent sides ($PQ = PS$) is a **rhombus**.
5. Since diagonals of a rhombus are perpendicular, $PR \perp QS$.

Is the converse true?

Yes. If $PR \perp QS$, the diagonals of parallelogram $PQRS$ are perpendicular, making $PQRS$ a rhombus ($PQ = PS$). Since $PQ = \frac{1}{2} AC$ and $PS = \frac{1}{2} BD$, this implies $AC = BD$.

Question 5

Suppose $PQRS$ is the Varignon parallelogram of $ABCD$.
(i) Copy only $PQRS$ on another paper. Show how you will recreate a congruent copy $A'B'C'D'$ of $ABCD$ from $PQRS$.
(ii) There are multiple ways to construct $A'B'C'D'$ in (i). Justify why the quadrilateral $A'B'C'D'$ you constructed is congruent to $ABCD$.
(iii) Show that if $PQRS$ is a square, then AC and BD are perpendicular and equal. Prove the converse.

SOLUTION:

(i) Construction Steps:

1. Pick any arbitrary point A' anywhere on the paper.
2. Draw a line from A' through P and extend it to point B' such that $A'P = PB'$ (making P the midpoint of $A'B'$).
3. From B', draw a line through Q and extend it to C' such that $B'Q = QC'$ (making Q the midpoint of $B'C'$).
4. From C', draw a line through R and extend it to D' such that $C'R = RD'$ (making R the midpoint of $C'D'$).
5. Connecting D' back to A' automatically ensures S is the midpoint of $D'A'$.
6. The resulting quadrilateral $A'B'C'D'$ is a congruent copy of $ABCD$.

(ii) Justification:

By construction, P, Q, R, S are the midpoints of sides $A'B', B'C', C'D', D'A'$. Any choice of initial point A' produces a quadrilateral $A'B'C'D'$ whose sides and diagonals are fixed relative to parallelogram $PQRS$. By SAS triangle congruences on the corner triangles ($\Delta SD'R \cong \Delta SDR$, etc.), all corresponding sides and angles of $A'B'C'D'$ match those of $ABCD$, proving $A'B'C'D' \cong ABCD$.

(iii) Proof:

- **If $PQRS$ is a square:**
  1. Square sides are equal $\Rightarrow PQ = PS \Rightarrow \frac{1}{2} AC = \frac{1}{2} BD \Rightarrow AC = BD$.
  2. Square sides are perpendicular $\Rightarrow PQ \perp PS$. Since $PQ \parallel AC$ and $PS \parallel BD$, it follows that $AC \perp BD$.
  3. Thus, diagonals AC and BD are **perpendicular and equal**.
- **Converse (Given $AC = BD$ and $AC \perp BD$):**
  1. $AC = BD \Rightarrow PQ = PS$ (adjacent sides of $PQRS$ are equal).

2. $AC \perp BD \Rightarrow PQ \perp PS$ (adjacent sides meet at $90°$).
3. A parallelogram with equal and perpendicular adjacent sides is a **square**. Hence, $PQRS$ is a square.